

SPECIALIZED AMINO ACID METABOLISM

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INTRODUCTION

Amino Acid Catabolism

- ▶ Catabolism of amino acids means breaking down amino acids to obtain energy or to make other substances.

Step 1: Remove the amino group

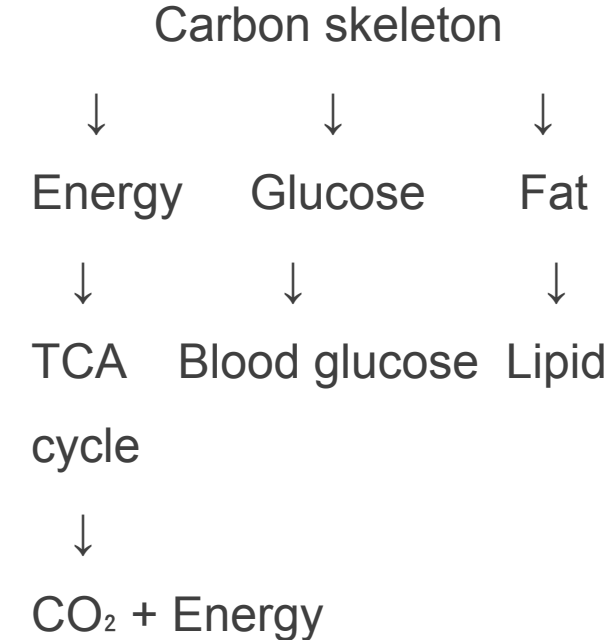
- ▶ Amino acid---Removal of α -amino group---Carbon skeleton remains

Step 2: Carbon skeleton enters metabolism

- ▶ The carbon skeletons are converted into 7 major products:

*Oxaloacetate	*Pyruvate
* α -Ketoglutarate	*Fumarate
*Succinyl CoA	*Acetyl CoA *Acetoacetate

- ▶ They can be used for:



- ▶ So, amino acids can ultimately be used to produce energy, make glucose, or make fat.

Essential vs Non-essential Amino Acids

Essential amino acids

- ▶ The Essential Amino Acids cannot be synthesized (or produced in sufficient amounts) by the body and therefore, must be obtained from the diet in order for normal protein synthesis to occur.

e.g: Cysteine, Tyrosine, Histidine, Valine.

Essential amino acids



Diet



Normal protein synthesis

Nonessential amino acids

- ▶ Non-Essential Amino Acids can be synthesized in sufficient amounts from the intermediates of metabolism or as in the case of cysteine and tyrosine from essential amino acids.

Metabolic intermediates



Nonessential amino acids

Examples:

- ▶ Cysteine → made from methionine
- ▶ Tyrosine → made from phenylalanine

Memorize the Essential Amino Acids

P.v.t. T.I.M. H.A.L.L.

Phe = Phenylalanine

Val = Valine

Thr = Threonine

Trp = Tryptophan

Ile = Isoleucine

Met = Methionine

His = Histidine

Arg = Arginine*

Leu = Leucine

Lys = Lysine

	Glucogenic	Glucogenic and Ketogenic	Ketogenic
Nonessential	Alanine Arginine* Asparagine Aspartate Cysteine Glutamate Glutamine Glycine Proline Serine	Tyrosine	
Essential	Histidine Methionine Threonine Valine	Isoleucine Phenylalanine Tryptophan	Leucine Lysine

*Arg is semiessential, depending on developmental stage

A. Glucogenic vs Ketogenic Amino Acids

A. Glucogenic Amino Acids

- ▶ that are broken down into substances that can be used to make glucose.
- ▶ produces pyruvate or TCA cycle intermediates.

Glucogenic amino acid



Pyruvate or TCA intermediate



Gluconeogenesis



Glucose (Glucogenic = can make glucose)

- ▶ This mainly occurs in the liver and kidneys

B. Ketogenic Amino Acids

- ▶ that are broken down to produce acetoacetate, acetyl-CoA, or aceto acetyl-CoA.
- ▶ these substances can be used to make ketone bodies, rather than glucose.

Ketogenic amino acid



Acetyl-CoA / Aceto acetyl-CoA



Acetoacetate



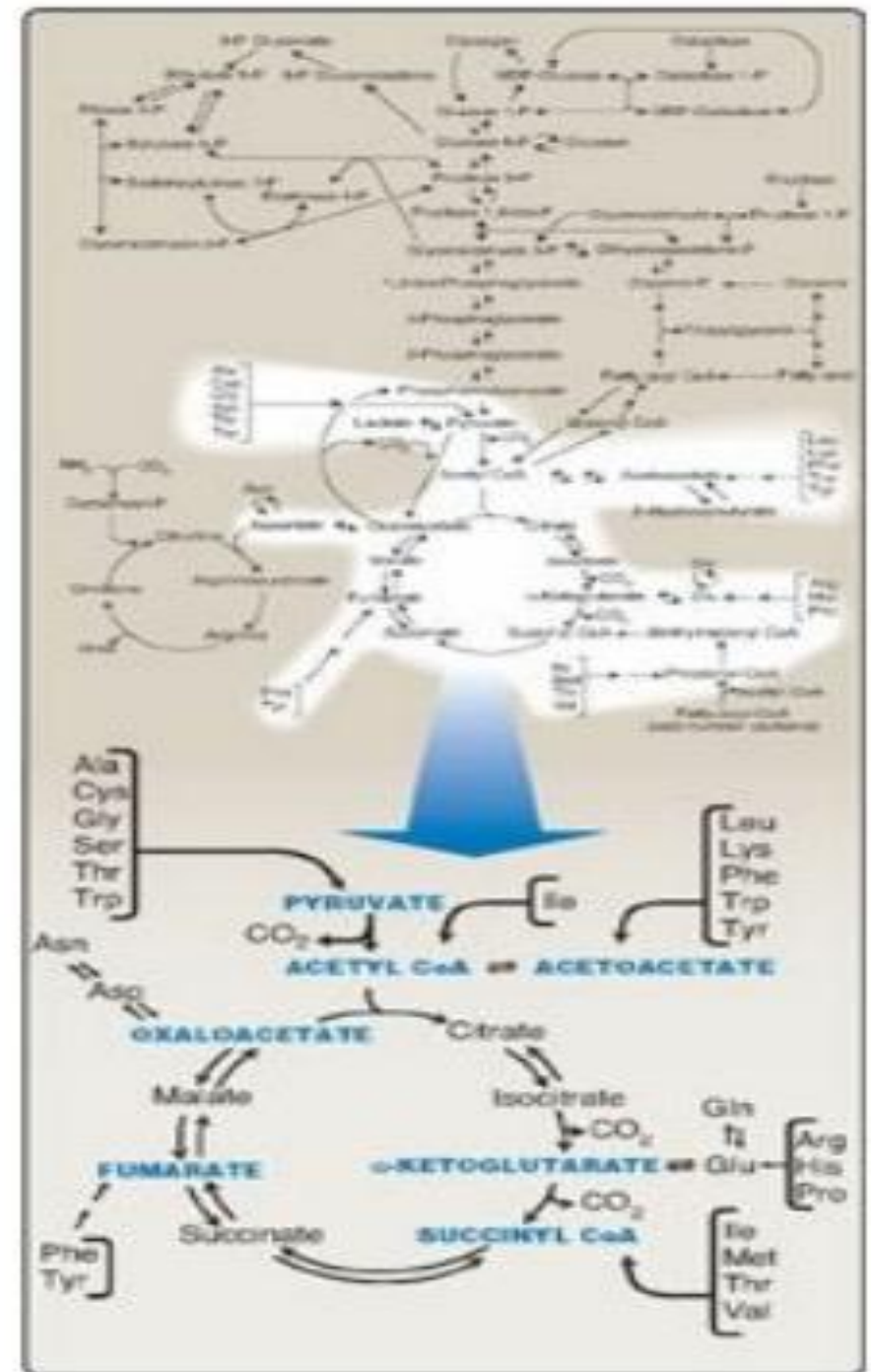
Ketone bodies



Energy production

Ketogenic Amino Acids

- ▶ The 3 ketone bodies are:
 - Acetoacetate
 - 3-Hydroxybutyrate
 - Acetone
- ▶ Leucine and lysine are the ONLY exclusively ketogenic amino acids.
 - Leucine → Ketogenic only
 - Lysine → Ketogenic only
- ▶ Their carbon skeletons cannot be converted into glucose through gluconeogenesis.

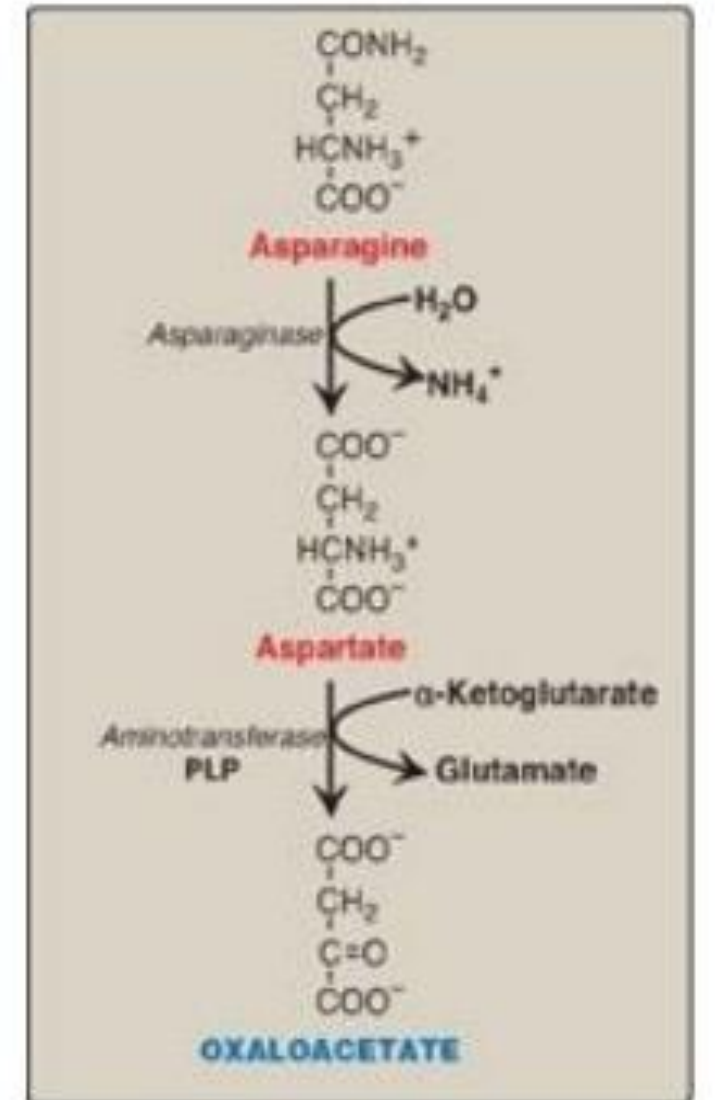


Catabolism of Carbon Skeleton & Amino Acids

- ▶ Amino acids that form Oxaloacetate
- ▶ Amino acids that form α -Ketoglutarate
- ▶ Amino acids that form Pyruvate
- ▶ Amino acid that form Fumarate
- ▶ Amino acid that form Succinyl CoA
- ▶ Amino acid that form Acetyl CoA or Aceto acetyl CoA

A . Amino Acids That Form Oxaloacetate

- ▶ Asparagine is hydrolyzed by asparaginase, liberating ammonium (NH_4^+) and aspartate
- ▶ Aspartate loses its amino group by transamination to form oxaloacetate
- ▶ Leukemic cells are unable to synthesize sufficient asparagine to support their growth. This makes asparagine an essential amino acid for these cells which therefore require asparagine from the blood.
- ▶ Asparaginase which hydrolyzes asparagine to aspartate, can be administered systemically to treat leukemia patients.
- ▶ It lowers the level of asparagine in the plasma and therefore, deprives cancer cell of a required nutrient.



B. Amino Acids That Form α -Ketoglutarate

1. Glutamine

Glutaminase

□ Glutamine ----- Glutamate

Transamination/Oxidative Deamination

□ Glutamate ----- α -Ketoglutarate

Glutamate dehydrogenase

2. Proline

Oxidation

□ Proline ----- Glutamate

□ Glutamate ----- α -Ketoglutarate

3. Arginine

Arginase

- Arginine ----- Ornithine (reaction occur in liver as part of urea cycle)
- Ornithine ----- α -Ketoglutarate

4. Histidine

HISTIDINE

↓ Histidase

UROCANIC ACID

↓

N-Formiminoglutamate (FIGlu)

↓ Tetrahydrofolate (THF)

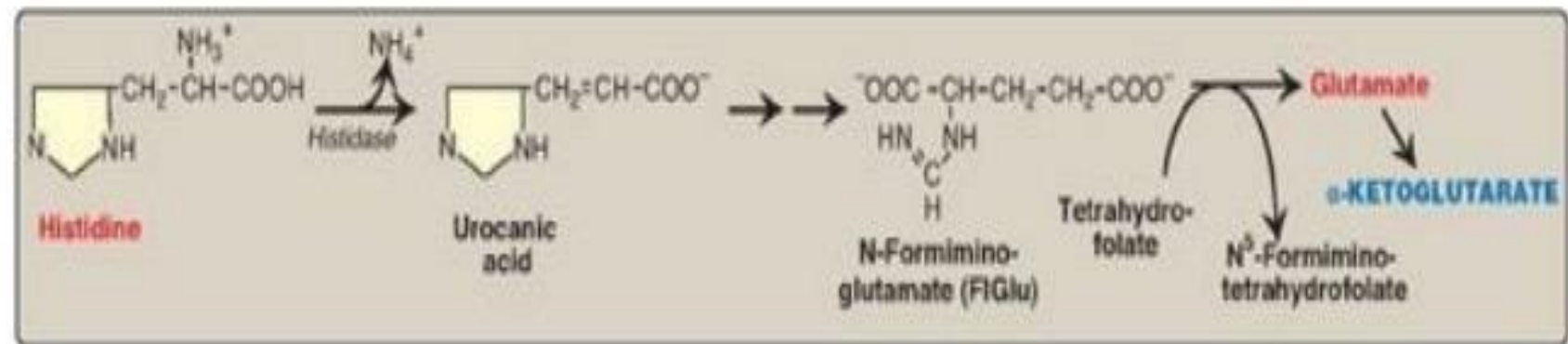
GLUTAMATE

+

N⁵-Formiminotetrahydrofolate

↓ Aminotransferase

α -KETOGLUTARATE



C . Amino Acids That Form Pyruvate

1. Alanine → Pyruvate

Alanine loses its amino group by transamination.

ALANINE ---- Transamination ---- PYRUVATE ----

Gluconeogenesis ---- GLUCOSE

- ▶ Alanine is the major gluconeogenic amino acid

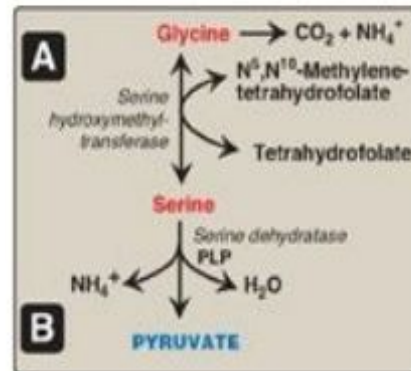
2. Serine

Pathway A: Serine → Glycine

- ▶ SERINE — Removal of hydroxymethyl group — GLYCINE + N⁵,N¹⁰-Methylene-THF

Pathway B: Serine → Pyruvate

- ▶ SERINE----Serine dehydratase
---- PYRUVATE
---- Gluconeogenesis
---- GLUCOSE



3. Glycine

Pathway A: Glycine ↔ Serine

GLYCINE + N⁵,N¹⁰-Methylene-THF



SERINE

This reaction is reversible.

Pathway B: Glycine → CO₂ + NH₄⁺

GLYCINE — Oxidation --- CO₂ + NH₄⁺

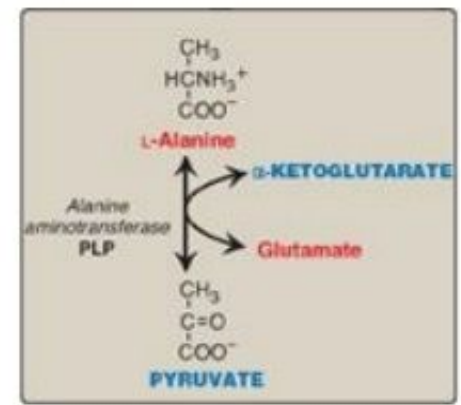
Pathway C: Glycine → Glyoxylate

GLYCINE — Deamination — GLYOXYLATE

↓ → Oxalate

→ Glycine

(by transamination)



4. Cystine:

- ▶ This amino acid undergoes desulfuration to yield pyruvate.

[Note: The sulfate released can be used to synthesize 3-phosphoadenosine-5I-phosphosulfate (PAPS), an activated sulfur donor to a variety of acceptors.]

Cysteine can be oxidized to its disulfide derivative, cystine.

NADH+

□ Cystine ----- Cysteine

Desulfuration

□ Cysteine ----- Pyruvate

5. Threonine:

- ▶ This amino acid is converted to pyruvate in most organisms but is a minor pathway (at best) in humans

□ Threonine ----- Pyruvate

□ Threonine ----- α-KetoButyrate

|

Succinyl-CoA

D. Amino Acids That Form Fumarate

Phenylalanine



Phenylalanine hydroxylase + BH_4



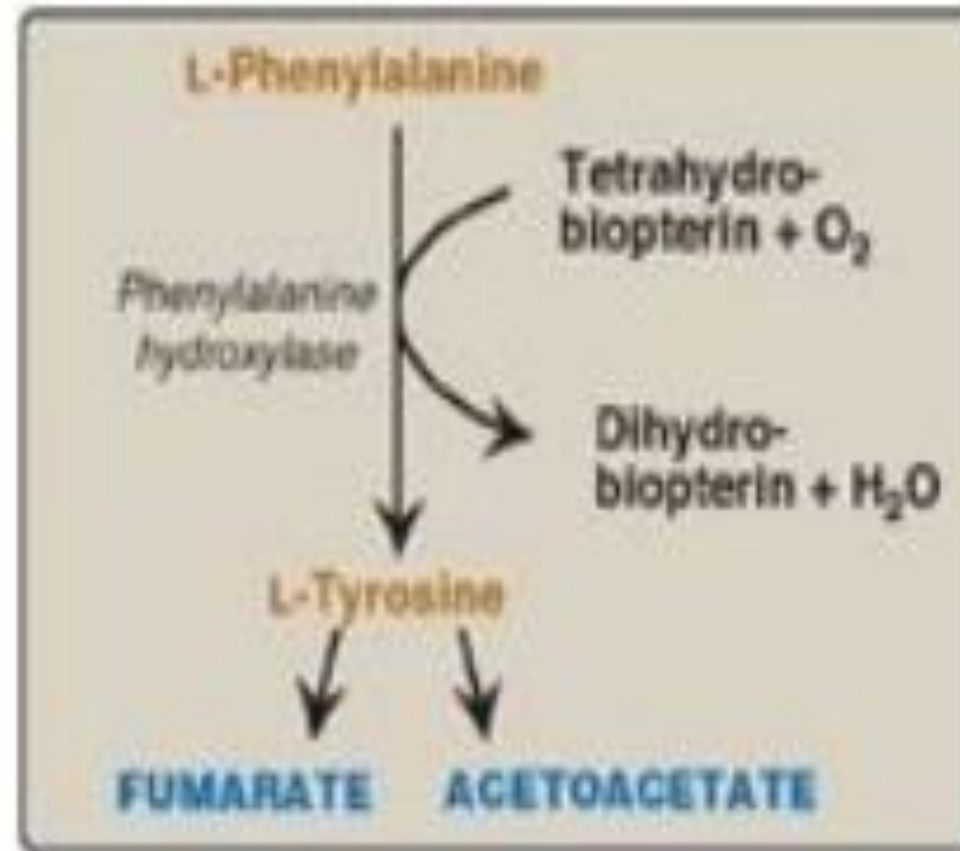
Tyrosine



Fumarate + Acetoacetate



Glucogenic + Ketogenic



E. Amino Acids That Form Succinyl CoA

1. Formation of S-adenosylmethionine(SAM)

First, methionine is activated by ATP.

METHIONINE + ATP

| Methionine adenosyltransferase



S-ADENOSYLMETHIONINE (SAM)

SAM is a high-energy compound.

Although ATP is used, SAM is unusual because SAM itself contains no phosphate.

The energy for the reaction effectively comes from hydrolysis of the three phosphate bonds of ATP.

2. SAM is the major methyl-group donor

SAM contains an activated methyl group ($-\text{CH}_3$) attached to sulfur.

SAM

| Activated CH_3



METHYL TRANSFER
REACTION



Acceptor molecule



Methylated product

Continued...

► Examples

SAM + Norepinephrine

| methyltransferase



Epinephrine

So:

- Norepinephrine + $\text{CH}_3 \rightarrow$ Epinephrine
- The methyl group can be transferred to:
- Nitrogen (N) $\rightarrow$ e.g. epinephrine synthesis
- Oxygen (O) $\rightarrow$ e.g. catechols
- Carbon (C) $\rightarrow$ e.g. cytosine

3. SAM becomes SAH

► After SAM donates its methyl group:

SAM



| loses CH_3



S-ADENOSYLHOMOCYSTEINE (SAH)

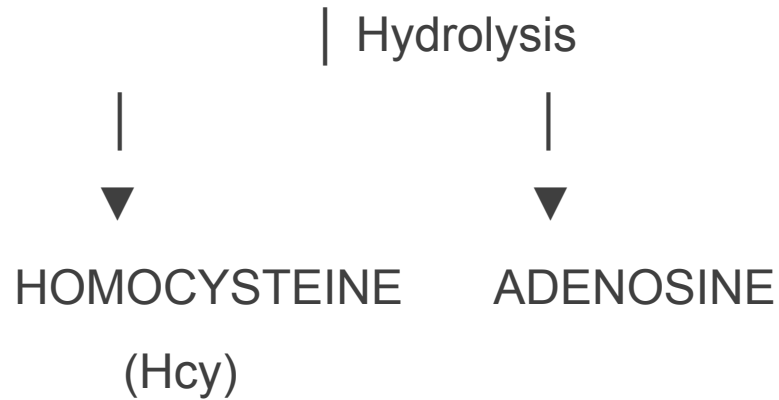
► So remember:

- SAM = methyl donor
- SAH = product after methyl donation

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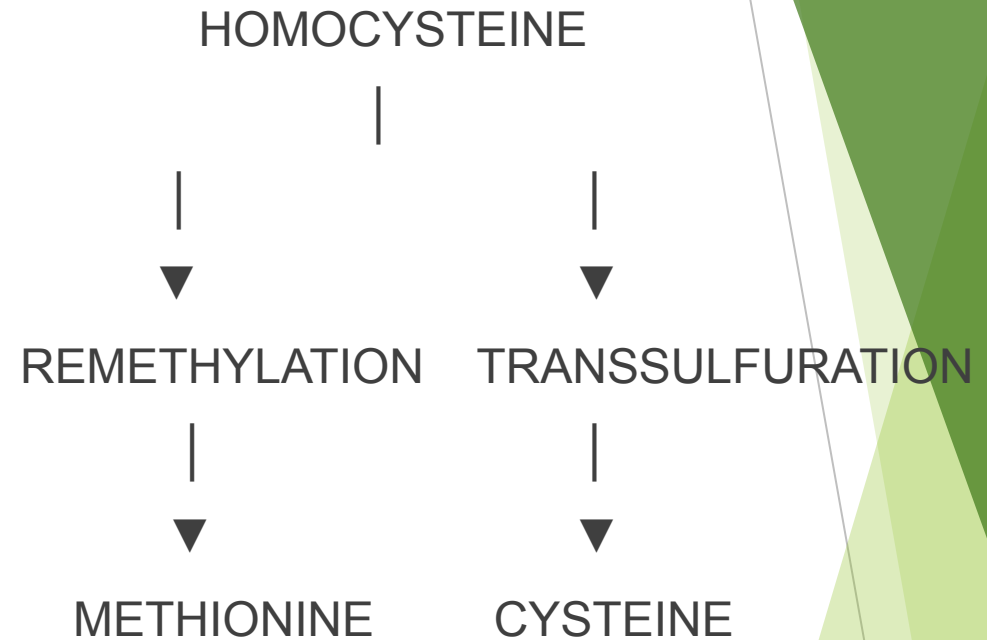
4. SAH is hydrolyzed

▶ S-ADENOSYLHOMOCYSTEINE (SAH)



- Now homocysteine is at the central branching point.
- It has two possible fates
- Homocysteine is a metabolite associated with atherosclerotic vascular disease

5. Homocysteine has TWO pathways



Continued...

A. Remethylation → Methionine

When the body needs to conserve/recycle methionine:

HOMOCYSTEINE

| + CH₃



METHIONINE

- ▶ Where does the methyl group come from?

- ▶ N⁵-methyl-THF

N⁵-methyl-THF

| supplies CH₃



Methylcobalamin (Vitamin B12)

| transfers CH₃



HOMOCYSTEINE



METHIONINE

Continued.....

- ▶ The enzyme is-Methionine synthase & it requires-Vitamin B12 (cobalamin)
- ▶ Simplified flow:

N⁵-methyl-THF

▼CH₃

Vitamin B12

▼CH₃

Homocysteine

▼Methionine synthase

Methionine

- ▶ So the key memory is
 - Homocysteine + methyl group → Methionine
 - Vitamin B12 is required

B. Transsulfuration → Cysteine

- ▶ The second pathway converts homocysteine into cysteine.

HOMOCYSTEINE

| + SERINE

▼

CYSTATHIONINE

|

▼

CYSTATHIONINE LYASE

| —→ CYSTEINE

▼

α-KETOBUTYRATE

- ▶ This pathway requires vitamin B6.

HOMOCYSTEINE + SERINE

| Vitamin B6



CYSTATHIONINE

| Vitamin B6



CYSTEINE

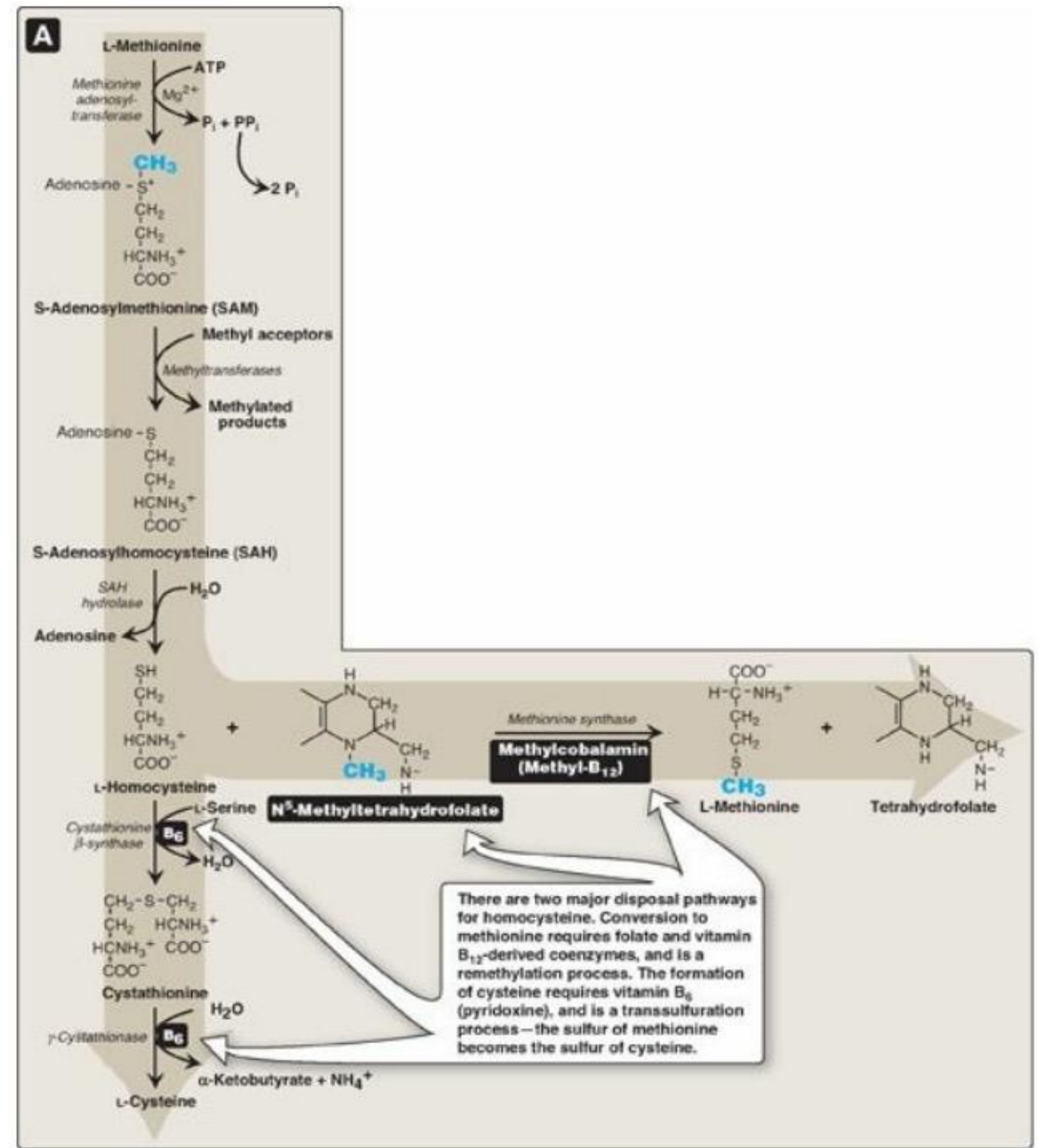
α-KETOBUTYRATE

▼ oxidative decarboxylation

PROPIONYL-CoA



SUCCINYL-CoA

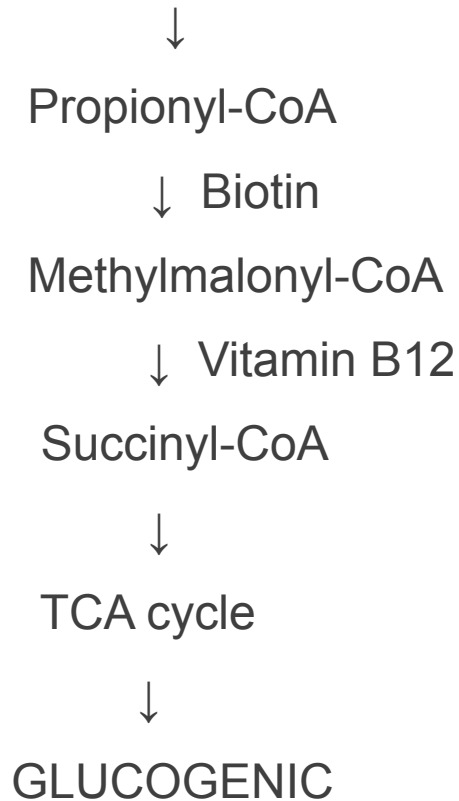


F. Other Amino Acids That Form Succinyl CoA

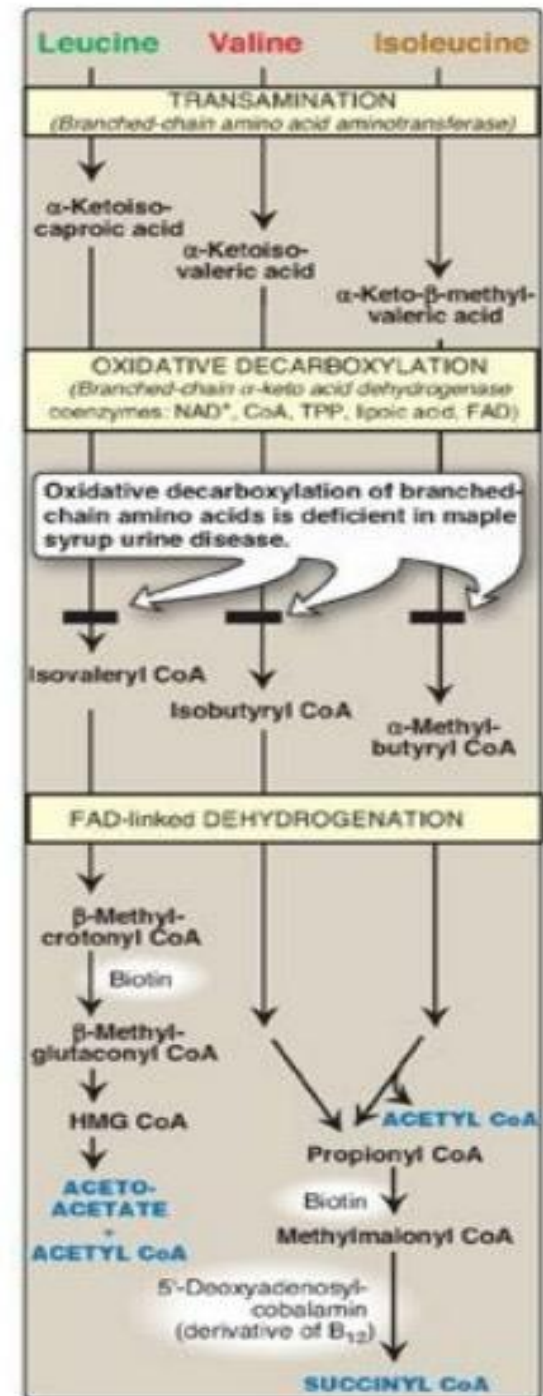
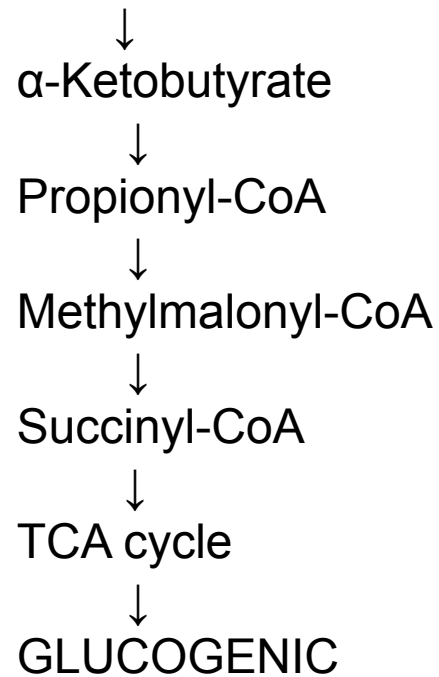
Valine, Isoleucine & Threonine

All three ultimately produce Succinyl-CoA, which enters the TCA cycle → therefore they are glucogenic.

VALINE + ISOLEUCINE

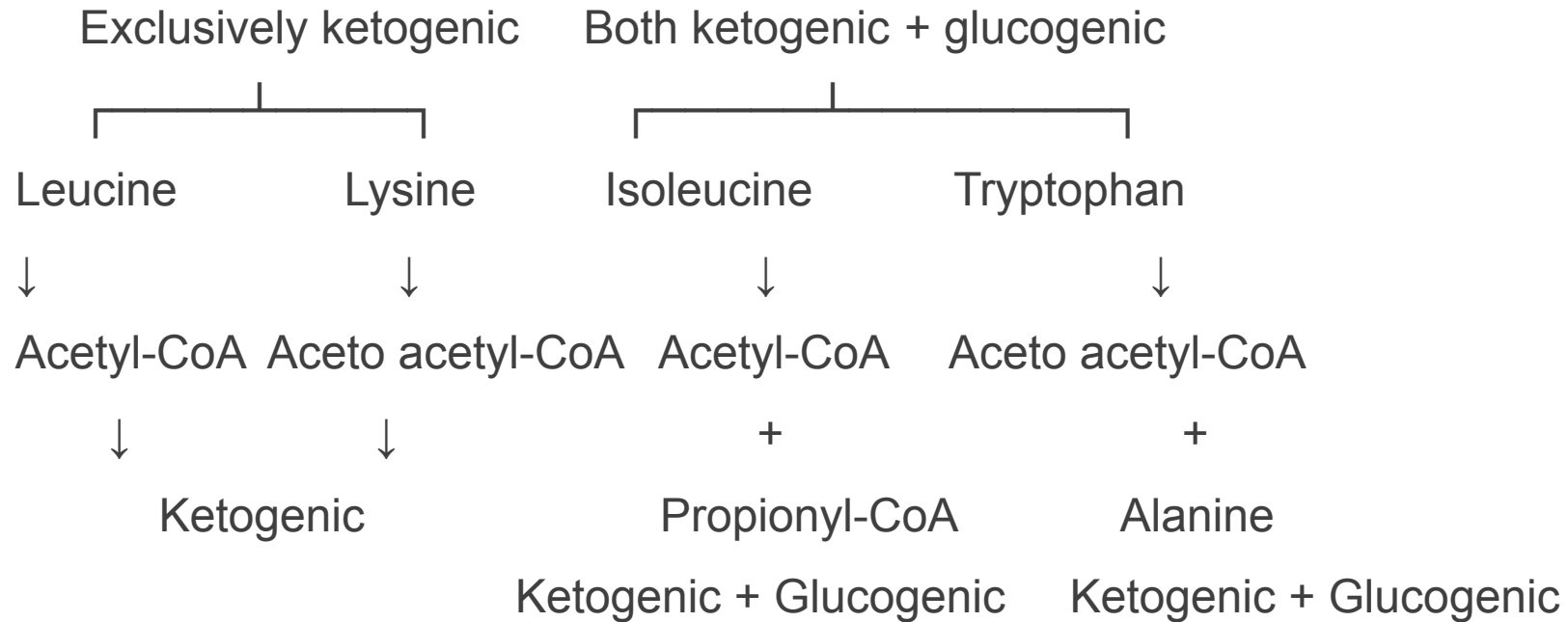


THREONINE



G. Amino Acids That Form Acetyl CoA Or Acetoacetyl CoA

KETOGENIC AMINO ACIDS



H. Catabolism Of The Branched-Chain Amino Acids

BCAAs: Leucine Isoleucine Valine



1. TRANSAMINATION

Enzyme: Branched-chain amino acid aminotransferase (BCAT)

Cofactor: Vitamin B6↓

α-Keto acids



2. OXIDATIVE DECARBOXYLATION

Enzyme: BCKD complex

Cofactors: TPP + Lipoic acid + FAD + NAD⁺ + CoA



Acyl-CoA



3. DEHYDROGENATION

FAD → FADH₂↓

α,β-unsaturated acyl-CoA

4. END PRODUCTS



LEUCINE



ISOLEUCINE



VALINE



Acetoacetate



Acetyl-CoA



SuccinylCoA

+ Acetyl-CoA + Succinyl-CoA



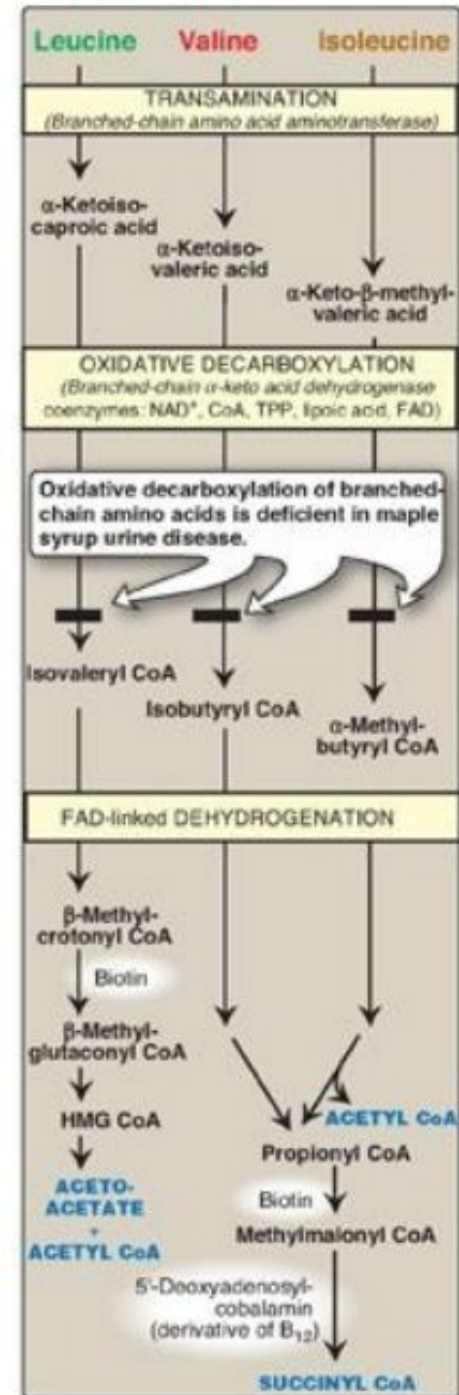
Ketogenic



Ketogenic +
Glucogenic



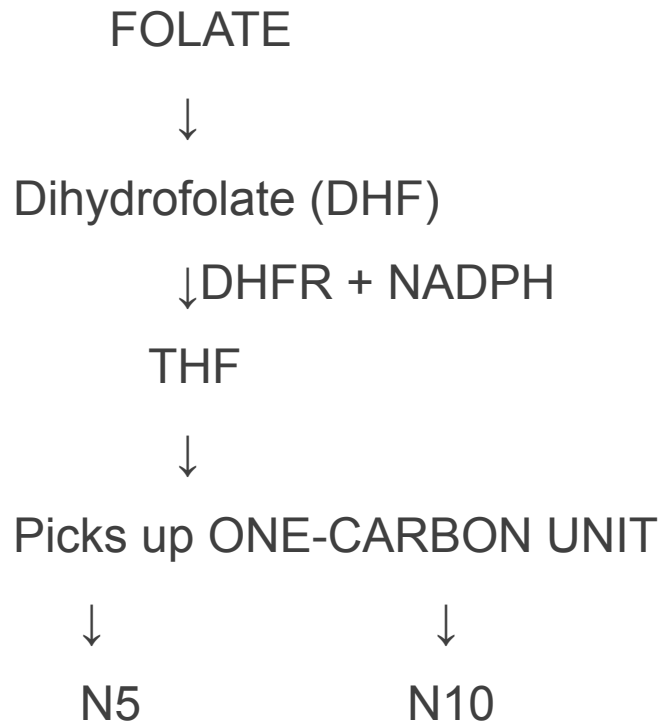
Glucogenic



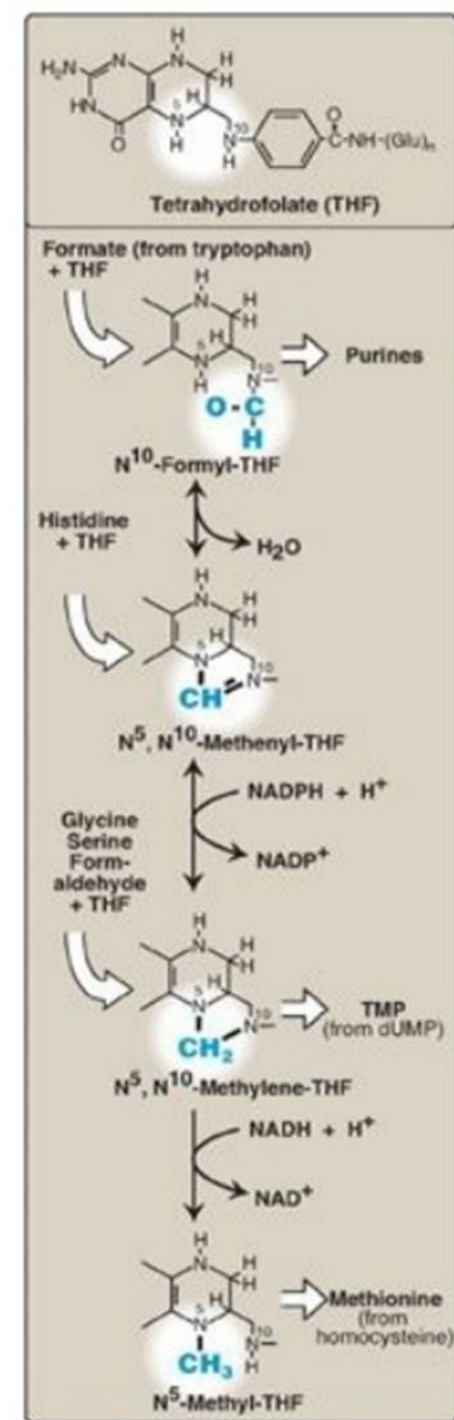
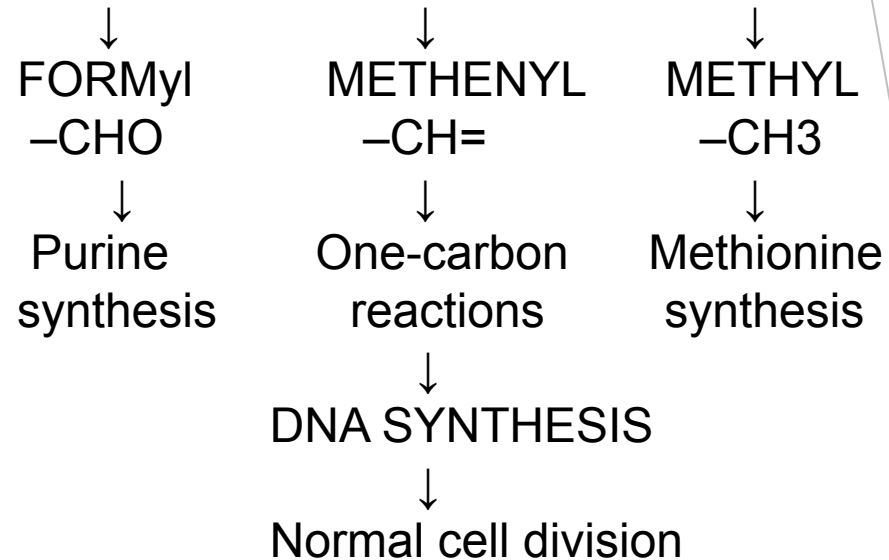
Folic & Amino Acids Metabolism

1. The one-carbon pool is a collection of single-carbon units (formyl, methenyl, methylene, and methyl) carried mainly by THF and SAM and transferred to molecules during synthesis.

2. Biotin carries CO_2 for carboxylation reactions but is not part of the one-carbon pool; defective biotin attachment causes multiple carboxylase deficiency, treated with biotin supplementation



ONE-CARBON POOL



BIOSYNTHESIS OF NON-ESSENTIAL AMINO ACID

A. Synthesis from α -keto acids

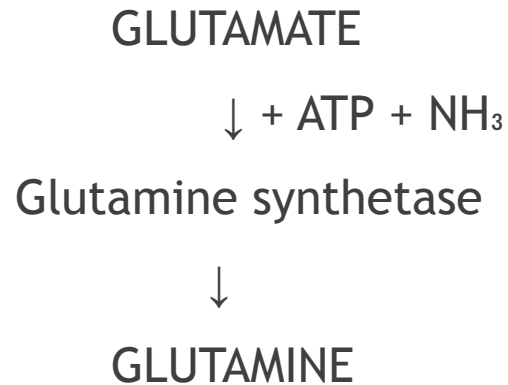
Nonessential amino acids are made from metabolic intermediates; for example, alanine, aspartate, and glutamate are formed by transferring an amino group to pyruvate, oxaloacetate, and α -ketoglutarate, respectively.

These reactions are called transamination; glutamate can also be formed from α -ketoglutarate and ammonia by glutamate dehydrogenase, especially when ammonia is high.

B. Synthesis by Amidation (Amidation means adding an amide group ($-\text{CONH}_2$) to a molecule using ammonia, usually with ATP energy)

1. Glutamine Synthesis

- ▶ Glutamine is made from glutamate by adding ammonia to its γ -carboxyl group.



Important:

Glutamate + NH_3 + ATP → Glutamine

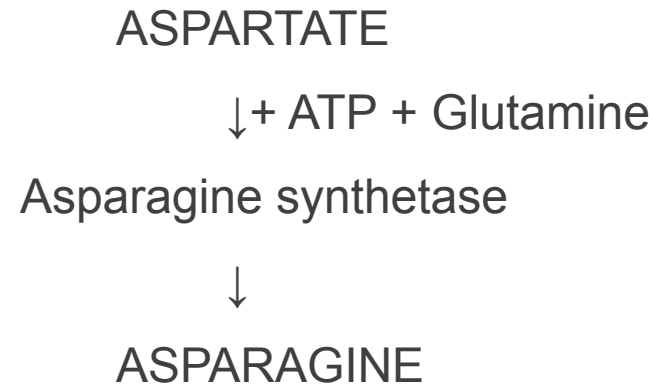
Glutamine synthetase + ATP → Glutamine

Used for protein synthesis

Carries ammonia safely in a nontoxic form

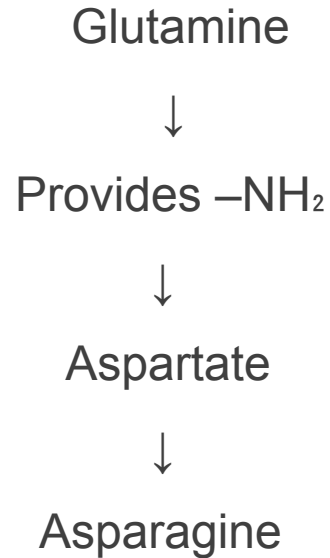
2. Asparagine Synthesis

Asparagine is made from aspartate, but the nitrogen/amide group is donated by glutamine.



Continued...

More specifically:

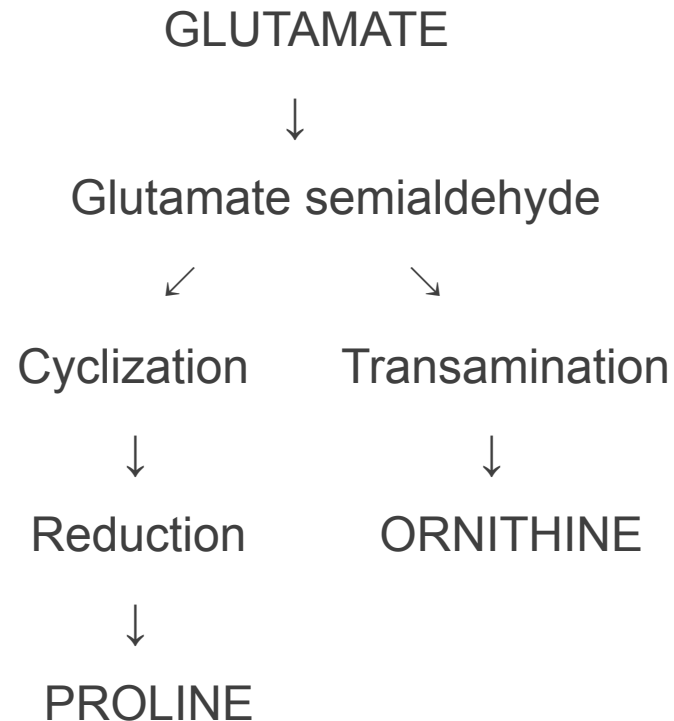


Asparagine synthetase + ATP + glutamine → Asparagine

- Important: The amide group is added to the β-carboxyl group of aspartate.

C. Proline Synthesis

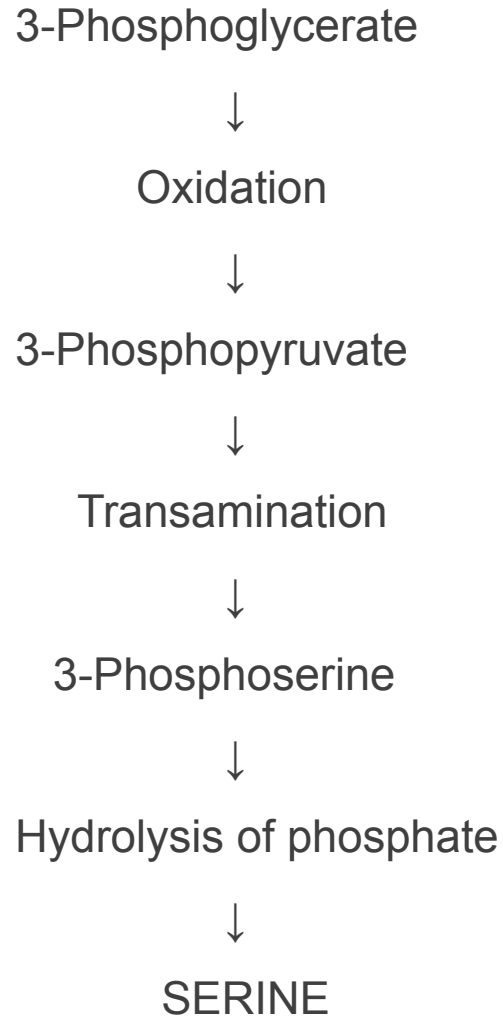
Proline is synthesized from glutamate through an intermediate called glutamate semialdehyde.



D. Serine, Glycine & Cysteine

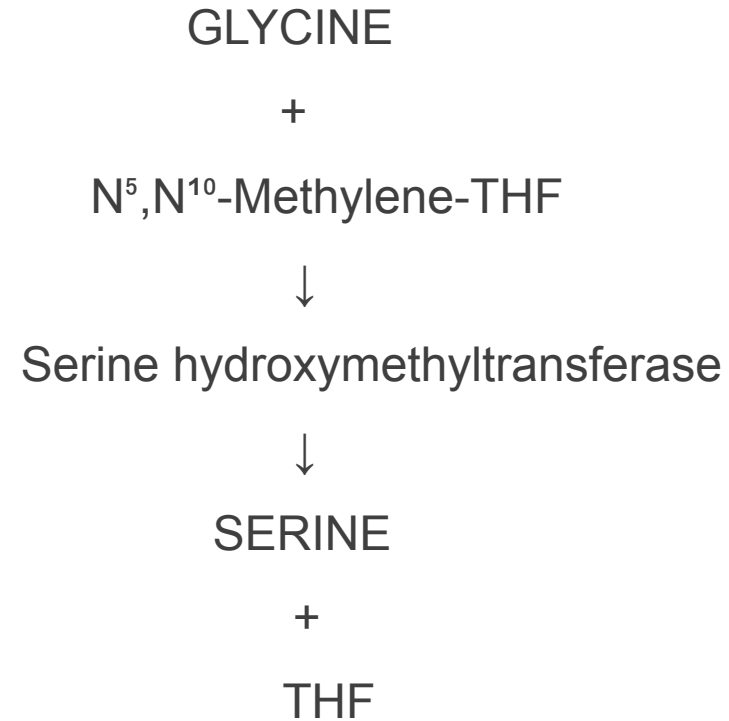
1. Serine Synthesis - From glycolysis

- ▶ Serine is synthesized from 3-phosphoglycerate, an intermediate of glycolysis.



Serine from Glycine

- ▶ Serine can also be produced from glycine.



Important: N⁵,N¹⁰-methylene-THF provides the hydroxymethyl group

Continued...

2. Glycine Synthesis

Glycine is synthesized from serine.

SERINE



Serine hydroxymethyltransferase



Removal of hydroxymethyl group



GLYCINE

- ▶ At the same time:

Hydroxymethyl group



THF Serine → Glycine + one-carbon unit → THF

3. Cysteine Synthesis

METHIONINE



Homocysteine + Serine



Cystathionine



Hydrolysis



Cysteine α-Ketobutyrate

- ▶ Cysteine is considered conditionally non essential because its synthesis depends on adequate methionine, which is an essential amino acid.

Continued

E. Tyrosine Synthesis

Tyrosine is synthesized from the essential amino acid phenylalanine.

PHENYLALANINE



Phenylalanine hydroxylase

↓ + O₂ + BH₄

TYROSINE

One oxygen atom becomes the –OH group of tyrosine.

The other oxygen atom becomes H₂O.

O₂ — O becomes –OH of TYROSINE

└ O becomes H₂O

BH₄ Recycling

During the reaction:

► BH₄-Oxidized-BH₂

BH₄ must then be regenerated:

► BH₂-Dihydropteridine reductase-NADH



BH₄ regenerated